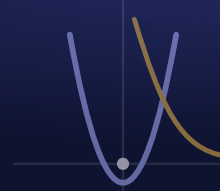


Principles of Applied Engineering

TEKS §127.781 • CTE Level 1 • Grade 9 • Mr. Gavaldon • Da Vinci School

robertgavaldon@burnhamwood.org


 SEMESTER 1
Week 1

MONDAY

Day 1 - Procedures, Syllabus & Your Design Notebook

Aug 10

OBJECTIVE

Students will set up a standards-compliant engineering design notebook (cover, table of contents, numbered and dated pages) and write a complete first entry with a labeled sketch and measurements recorded with correct units and tolerance, then evaluate a sample entry against a professional-documentation rubric.

TEKS

§127.781(d)(1) professional standards & employability; (d)(3)(B) maintain a design & computation engineering notebook

ELPS / EB SUPPORT

Model every notebook step live under the doc cam with icons and sentence stems (VC, CD); pre-teach notebook, entry, tolerance, and witness with picture cards (PV, DP). Give sentence frames for the integrity scenario ('The professional response is ___ because ___') and extra time on the units table (RS, ET). A bilingual peer is available for Google Classroom sign-in as backup, but the visual model carries it (PN).

AGENDA • 45 MIN

7' Bell-work: You invent something valuable; a competitor says they thought of it first. What evidence would prove it was you? (write in notebook)

10' Why engineers document: IP/legal, reproducibility, traceability, quality & safety (notes + notebook standards)

8' Anatomy of an entry + units/tolerance, modeled live under the doc cam

12' Set up notebook (cover, TOC, page numbers) + write a complete first entry with a labeled, dimensioned sketch

5' Rubric self-audit + Google Classroom sign-in

3' Exit ticket: 3 features that make a notebook legally defensible

SLIDES / ACTIVITY

The Engineer's Notebook: Your Professional Record

NOTEBOOK

Set it up today: cover, Table of Contents (pgs 1-3), number pages, date entries. First entry = what you hope to build this year.

TURN IN

Engineering Notebook - Setup, First Entry & Documentation Audit

OBJECTIVE

Students will distinguish engineering from science, identify at least four engineering disciplines with a historical milestone, map an engineered product's subsystems to the disciplines and career paths behind them, and produce an annotated technical sketch (isometric or orthographic) with labeled subsystems and dimensioned callouts.

TEKS

§127.781(d)(2)(A) engineering disciplines & their history; (d)(2)(E) engineering career paths; (d)(3)(C) sketching to present ideas

ELPS / EB SUPPORT

Guided-viewing sheet gives sentence stems and the video is paused at each discipline (RS, CD, WT). Pre-teach discipline, constraint, orthographic, and isometric with picture cards (PV, DP). Pair-share the science-vs-engineering question before whole-class share so EB students rehearse (PN, WT). A word bank supports the careers table (VC).

AGENDA • 45 MIN

6' Bell-work: list 5 engineered objects within reach; next to each, guess which kind of engineer made it

8' Science vs engineering + design under constraints (criteria/constraints/trade-offs)

12' Watch: Crash Course Engineering #1 + guided viewing (disciplines & history)

8' History of the disciplines + careers, pay, and pathways

8' Technical sketching mini-lesson + annotate an engineered object

3' Exit ticket

SLIDES / ACTIVITY

What Engineers Do: Disciplines, History & the Sketch That Communicates It

NOTEBOOK

Glue in the guided-viewing sheet. Sketch + label one engineered object; name its branch and the problem it solves.

TURN IN

Engineering Fields, Careers & Technical Sketch

OBJECTIVE

Students will apply the engineering design process to define a real-world problem and begin designing a solution (Ask, Imagine, Plan).

TEKS

§127.781(d)(2)(G) the engineering design process; (d)(1)(B) collaborate as a team

ELPS / EB SUPPORT

The design-report sheet is the scaffold, released one part at a time (CD, VC). Pre-teach criteria, constraint, variable, and iteration with gestures and quick examples (PV). Mixed-level teams of 2-3 with defined jobs so EB students use process vocabulary in context (PN). A sentence frame is provided for the justification (RS).

AGENDA • 45 MIN

5' Bell-work: purpose of the engineering design cycle

16' Present: the 5-step engineering design process

7' Pick a real problem (list 3, choose 1)

17' Start the design sheet: Ask -> Imagine (4 idea sketches) -> Plan (Design #1 + materials)

SLIDES / ACTIVITY

Engineering Design Process — Pick Your Problem

NOTEBOOK

Glue in the design-process record sheet. Sketch your two candidate designs before building.

TURN IN

Engineering Design Process sheet - Ask / Imagine / Plan

OBJECTIVE

Students will finish their final design, write a structured AI prompt, and generate + submit a product image.

TEKS

§127.781(d)(2)(G) engineering design process – iteration & constraints; (d)(1)(B) teamwork

ELPS / EB SUPPORT

Reuse Day 3's process vocabulary plus load, compression, buckling, and triangulation with a quick physical demo (repetition builds language: RS, PV, DP). Roles (structural lead / builder / tester-recorder) give every student a clear job (CD). The efficiency and percent-improvement formulas are shown worked before students compute (VC, ET).

AGENDA • 45 MIN

8' Bell-work + video (start 1:28): name 2 improvements and why each matters

10' Model the notebook page (prompt on top, sketch on bottom) + copy the prompt

20' Finish final design, then generate the product image in Gemini

7' Submit the image to Google Classroom

SLIDES / ACTIVITY

Finish Your Design + Product Image

NOTEBOOK

Log Build 1 and Build 2 heights. Note the one change you made and why. Sketch the strongest shape you found.

TURN IN

AI product image (Gemini) + notebook page (prompt + final design) -> Google Classroom

OBJECTIVE

Students will classify products by engineering discipline, analyze a technology's benefits, trade-offs, and unintended consequences on society, evaluate it with a weighted impact matrix, and defend an adopt/redesign/restrict recommendation with evidence.

TEKS

§127.781(d)(2)(A) fields of engineering; (d)(5) impact of technology on society

ELPS / EB SUPPORT

The impact matrix uses a word bank and numeric ratings so language load stays low while thinking stays high (VC, CD). Pre-teach discipline names, sustainability, trade-off, and unintended consequence with images (PV, DP). Team scoring lets EB students reason with a peer before writing (PN, WT). A sentence frame supports the recommendation (RS).

AGENDA • 45 MIN

5' Bell-work: match 4 products to their engineering discipline and justify one

10' Branches deep-dive + emerging/interdisciplinary fields + systems thinking

8' Impact of technology: benefits, trade-offs, unintended consequences, life-cycle, ethics

12' Team analysis: score a technology on a weighted impact matrix + defend a recommendation

7' Spot-the-engineering audit (room/campus) with discipline + trade-off

3' Exit ticket

SLIDES / ACTIVITY

Branches of Engineering & Their Impact on Society

NOTEBOOK

Record your 3 campus finds: what it is, its branch, and the problem it solves.

TURN IN

Technology Impact Analysis + Discipline Audit